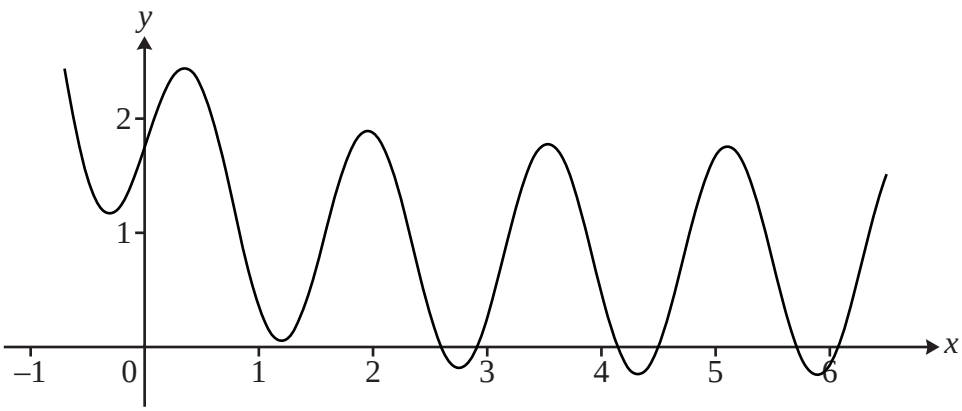
**8** Kareem wants to solve the equation $\sin 4x + e^{-x} + 0.75 = 0$. He uses his calculator to create the following table of values for $f(x) = \sin 4x + e^{-x} + 0.75$.

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $f(x)$ | 1.750 | 0.361 | 1.875 | 0.263 | 0.480 | 1.670 | −0.153 |

He argues that because $f(6)$ is the first negative value in the table, there is a root of the equation between 5 and 6.

**(a)** Comment on the validity of his argument. **[1]**

The diagram shows the graph of $y = \sin 4x + e^{-x} + 0.75$.




**(b)** Explain why Kareem failed to find other roots between 0 and 6. **[1]**

Kareem decides to use the Newton-Raphson method to find the root close to 3.

**(c) (i)** Determine the iterative formula he should use for this equation. **[2]**

**(ii)** Use the Newton-Raphson method with $x_0 = 3$ to find a root of the equation $f(x) = 0$. Show **three** iterations and give your answer to a suitable degree of accuracy. **[3]**

Kareem uses the Newton-Raphson method with $x_0 = 5$ and also with $x_0 = 6$ to try to find the root which lies between 5 and 6. He produces the following tables.

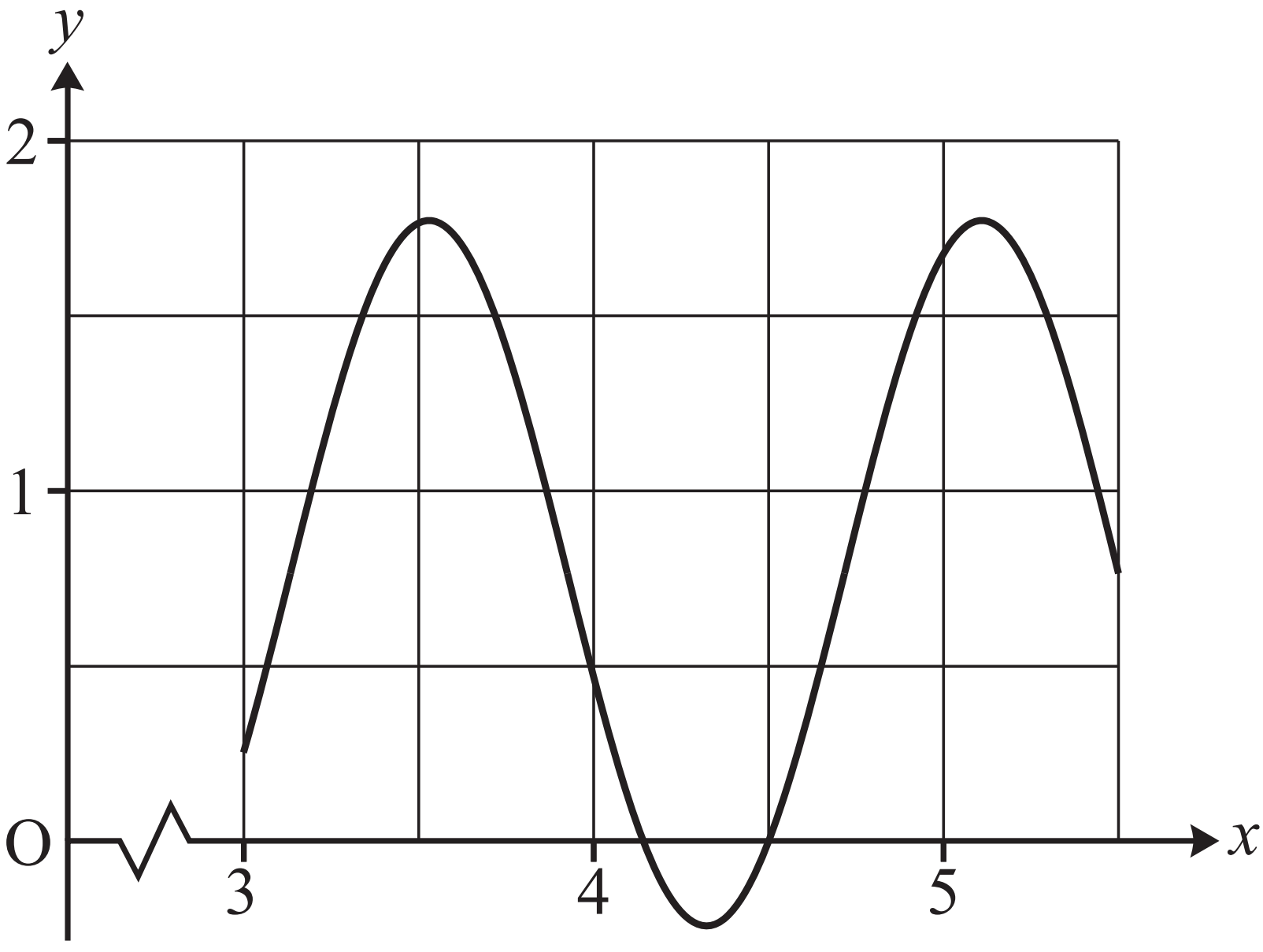
| $x_0$ | 5 |
|---|---|
| $x_1$ | 3.97288 |
| $x_2$ | 4.12125 |

| $x_0$ | 6 |
|---|---|
| $x_1$ | 6.09036 |
| $x_2$ | 6.07110 |

**(d) (i)** For the iteration beginning with $x_0 = 5$, represent the process on the graph in the Printed Answer Booklet. **[2]**

**(ii)** Explain why the method has failed to find the root which lies between 5 and 6. **[2]**

**(iii)** Explain how Kareem can adapt his method to find the root between 5 and 6. **[1]**


y
2
1
O
3
4
5
x